

A shared item representation forces a three-way trade-off in neural IRT
ability overfits wide, discrimination needs wide, difficulty is content-narrow

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Setup: amortized neural IRT

- Encoder (LSTM): response history $\rightarrow$ ability θ_t
- Item embedding $\rightarrow$ discrimination α_i , difficulty (step thresholds) β_i ; GPCM decoder

$$P(Y_i = k \mid \theta) = \frac{\exp \sum_{m=1}^k \alpha_i(\theta - \beta_{i,m})}{\sum_{l=0}^{K-1} \exp \sum_{m=1}^l \alpha_i(\theta - \beta_{i,m})} \quad (2\text{PL at } K=2)$$

- Trained on response prediction only. No supervision on θ, α, β .
- Scale is gauge-free $\Rightarrow$ recovery = rank (Spearman ρ).

The item code has two consumers

- a **value** embedding v_q (narrow) $\rightarrow$ encoder input $\rightarrow$ **ability** θ
 - a **key** embedding k_q (wide) $\rightarrow$ **discrimination** α , **difficulty** β readouts
 - **Shared**: one embedding, width W , feeds both consumers
 - **Decoupled**: a narrow value embedding + a separate wide key embedding
- $\Rightarrow$ *shared ties the encoder-input width to the readout width.* That tie is the problem.

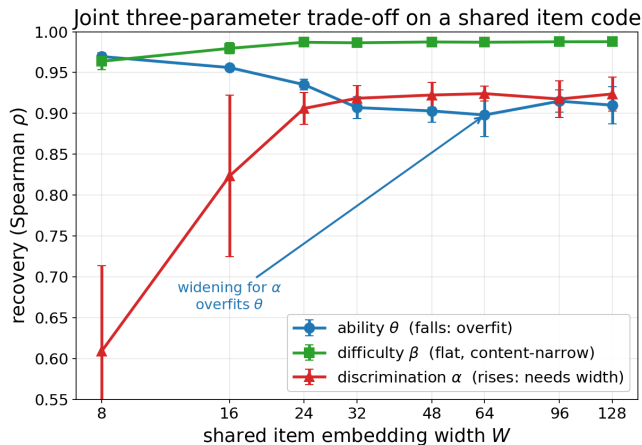
Three parameters, three opposite preferences

Fisher information ($w = p(1-p)$):

$$I(\theta) = I(\beta) = \alpha^2 w, \quad I(\alpha) = (\theta - \beta)^2 w$$

- α **wants the readout WIDE**, low Fisher ($\rightarrow 0$ at $\theta=\beta$), needs capacity to be read off
- θ **wants the encoder input NARROW**, a wide input lets the encoder absorb item idiosyncrasy and overfit ability
- β **is content NARROW**, high Fisher, read directly, no pooling; needs no width and is not hurt by it

The phenomenon: a three-way trade-off



Widen the shared code W :

- α rises 0.61 $\rightarrow$ 0.92 (gets capacity)
- θ falls (overfits the wide input)
- β flat-high 0.96 $\rightarrow$ 0.99

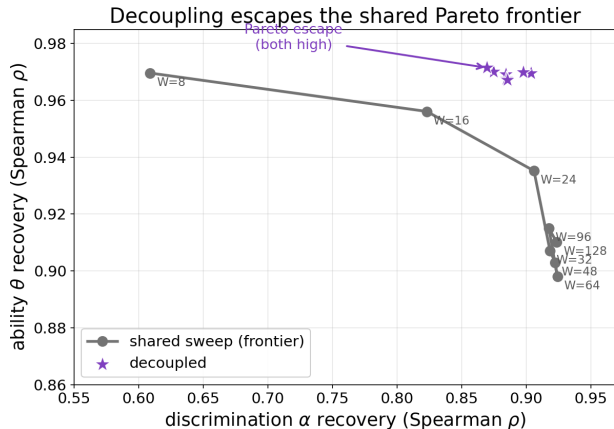
One knob, opposed pressures. β does not pay.

Two trade-offs of different KIND (the key point)

- α : **a RATE effect, endpoint invariant.** α is a free per-item table, so the optimum is unbiased (free-table invariant). Sharing only slows it (conditioning κ); decoupling speeds it. Transient: vanishes as training $\rightarrow \infty$, not at a fixed budget. (P9)
- θ : **an ENDPOINT effect, not invariant.** θ is read by the amortized encoder, NOT a free table. A wide input overfits it, $\text{Var}(\hat{\theta}) \sim \sigma^2 W/n$. A finite-data generalization loss that does NOT vanish at convergence. (P10)
- β : **indifferent control.** High Fisher, non-pooled $\Rightarrow$ neither needs width nor is hurt ($\Delta_\beta \approx 0$). (P11)

Not one phenomenon. A slow mode (rate) *and* an overfit (endpoint), tied by one width.

Not capacity: allocation (the gate)

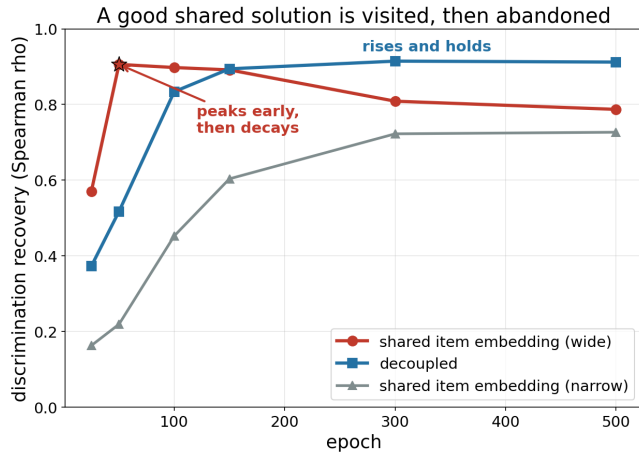


Control: matched total item-embedding params.

- shared sweep traces a Pareto frontier
- decoupled sits **above** it, even at $6\times$ the shared budget

$\Rightarrow$ allocation, not budget. Decoupling escapes the frontier (P12).

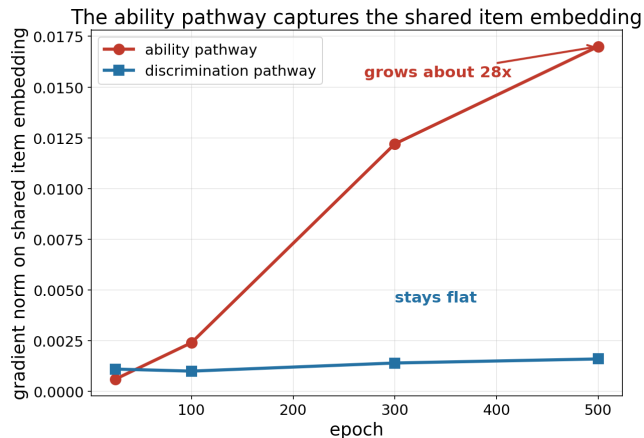
α side, dynamics: reached then left



- wide-shared α : peaks 0.91 (ep 50), decays to 0.79 (ep 500)
- decoupled: rises 0.91, holds

$\Rightarrow$ the good α solution is reachable, reached, then abandoned = dynamics, not a ceiling.

α side, mechanism: the shared code is conditioned



- θ -pathway pull grows $\sim 28\times$;
 α -pathway flat
- $\cos(g_\theta, g_\alpha) \approx 0$ (not a tug-of-war)
- the lag is eigenvalue spread,
stiffness $\kappa \approx 1.9$ at $K=4$

α side, the gradient-flow ODE

One item's embedding e_q . Near its optimum, $\delta = e_q - e_q^*$, H_q the Hessian in e_q 's directions:

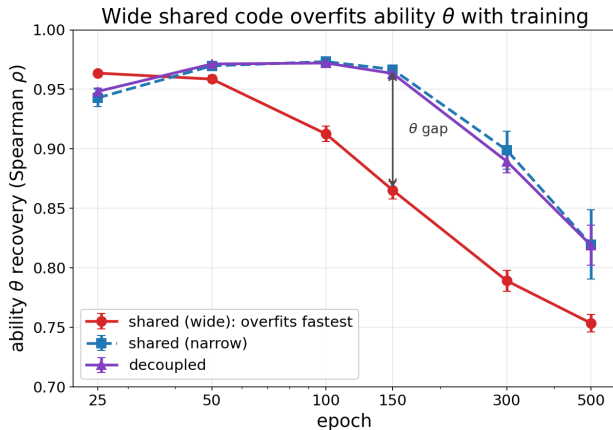
$$\dot{\delta} = -H_q \delta, \quad \delta_c(t) = \delta_c(0) e^{-\lambda_c t}.$$

- Shared: e_q feeds encoder (θ) and α, β readouts $\Rightarrow$ steep θ direction ($\sim I(\theta)$), flat α direction ($\sim I(\alpha)$): $\kappa = I(\theta)/I(\alpha) \gg 1$
- one step $\eta < 2/\lambda_{\max}$: slow mode $\sim 1/\kappa \Rightarrow \tau_{\alpha}^{\text{sh}} \sim O(\kappa)$; decoupled $O(1)$

speedup $\sim \kappa = I(\theta)/I(\alpha)$ (grows with K)

Same fixed point $\Rightarrow$ rate, not endpoint. Largest under GD; Adam compresses κ .

θ side, overfitting: an endpoint effect



- wide-shared θ overfits worst: $0.96 \rightarrow 0.75$
- decoupling keeps the encoder narrow $\Rightarrow$ protects θ at the operating point

Bias and variance: $\text{Var}(\hat{\theta}) \sim \sigma^2 W/n$. An **endpoint** loss (not a rate), outside the free-table invariant. (The decay with long training, even narrow, is a separate regularization issue, not the width effect.)

The resolution: untie the two widths

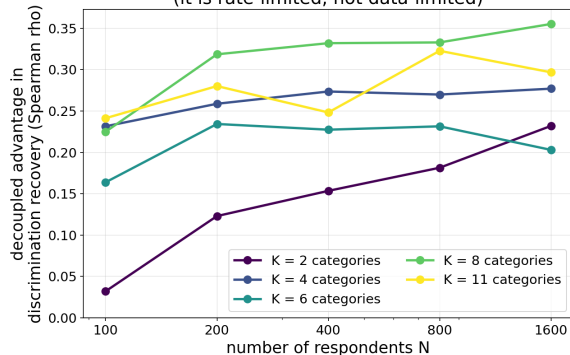
- **Shared:** one width, opposed knobs ($d\alpha/dW > 0$, $d\theta/dW < 0$, $d\beta/dW \approx 0$) $\Rightarrow$ a Pareto frontier, no W wins both
- **Decoupled:** two independent widths $\Rightarrow$ narrow encoder (good θ, β) + wide readout (good α); each optimized by its own knob

$\Rightarrow$ decoupling is Pareto-**dominant**: it escapes the frontier, not moves along it. (P12)

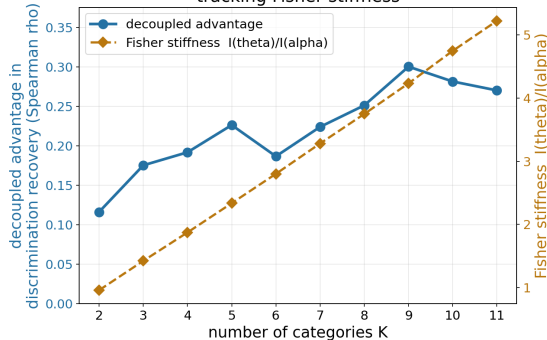
It cures both pressures at once: θ keeps its bottleneck, α gets its capacity.

The α rate effect: finite-sample and K -scaled

More data does not shrink the advantage
(it is rate-limited, not data-limited)

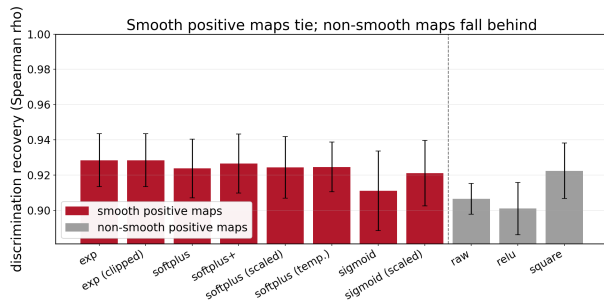


The advantage grows with the number of categories K ,
tracking Fisher stiffness



- Fixed budget, vary N : the decoupling advantage does NOT shrink with data (rate-limited).
- Across $K=2$ to 11: it tracks the Fisher stiffness κ , $\rho = 0.89$.

Ruled out: the positivity map



$\alpha = g(\cdot)$, g positive. Is $g = \exp$ special?

- matched init + per-map LR: exp ties all smooth maps ($\Delta \approx 0$)
- direct- α control (θ, β frozen): scalar preconditioner does nothing ($\kappa=1$)

$\Rightarrow$ positivity needed, exp not special.

The lever is the representation.

Conclusion

A shared item code ties the encoder-input width to the readout width, forcing a three-way trade-off: θ overfits wide, α needs wide, β is content-narrow. Decoupling unties the widths and dominates.

- Two mechanisms of different kind: α a convergence-RATE effect (endpoint invariant, $O(\kappa)$, grows with K); θ an ENDPOINT overfit ($\text{Var} \sim W/n$). β indifferent.
- A measurement-validity condition: recovered parameters are otherwise representation-allocation artifacts.
- Positivity map ruled out as the lever.

Scope: synthetic GPCM, LSTM (architecture-independent on swap), 3 to 10 seeds. θ -overfit formalized for a linear amortizer (LSTM lift argued). Open: NRM slopes, GRM, real data.